PROJECT REPORT

ON

Comparative Evaluation of Machine Learning Models for Early Mental Health Risk Prediction using Workplace Survey Data

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"**Comparative Evaluation of Machine Learning Models for Early Mental Health Risk Prediction using Workplace Survey Data**"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shyam Goyal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Mental health disorders have become a growing global concern affecting individuals across different age groups and professional environments. Early identification of individuals who may be at risk of mental health problems is important for providing timely support and preventing severe psychological outcomes. This research proposes a machine learning based approach to predict mental health treatment needs using structured workplace survey data. The dataset includes demographic information, workplace conditions, and personal mental health history. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature selection were applied before model training. Four supervised machine learning algorithms were implemented: Logistic Regression, Support Vector Machine, Random Forest, and Gradient Boosting. The dataset was divided into 80% training data and 20% testing data. Model performance was evaluated using accuracy and F1 score. Experimental results indicate that ensemble learning models outperform traditional classification models. Gradient Boosting achieved the highest accuracy of 75.39% and an Fl score of 0.75. Feature importance analysis revealed that work interference due to mental health, family history of mental illness, and workplace care options are the most influential predictors. The results demonstrate that machine learning can support early mental health risk detection although it should not replace professional clinical diagnosis.

1. Introduction

Mental health disorders have become increasingly prevalent in recent years, affecting individuals across different age groups, professions, and social backgrounds. Factors such as rising stress levels, intense workplace pressure, changing lifestyles, and reduced work-life balance have significantly contributed to this growing concern. In many cases, individuals fail to recognize early symptoms or hesitate to seek help due to stigma or lack of awareness. As a result, conditions like anxiety, depression, and burnout often go unnoticed until they become severe. Therefore, early identification of individuals who may require mental health support is crucial, as it enables timely intervention, reduces long-term impact, and improves overall well-being.

With the rapid advancement of data science and artificial intelligence, machine learning has emerged as a powerful tool for analysing large amounts of data and uncovering hidden patterns. In the context of mental health, machine learning techniques can be applied to survey-based datasets that include information about personal history, workplace environment, and behavioural factors. By learning from such data, these models can identify relationships between different variables and predict the likelihood of an individual requiring mental health treatment.

This project aims to leverage these capabilities by building and evaluating multiple machine learning models for early prediction of mental health conditions. By comparing different algorithms and analysing their performance, the study seeks to develop a system that can act as a supportive tool for early detection. Such a system can help organizations, educational institutions, and healthcare providers take proactive measures, promote awareness, and encourage individuals to seek help at the right time.

2. Objectives

- To develop a machine learning model for predicting mental health conditions
- To preprocess and clean real-world survey data
- To apply and compare multiple classification algorithms
- To evaluate models using performance metrics
- To identify key factors influencing mental health conditions

3. Dataset Description

The dataset used in this project is sourced from Kaggle, which is “Mental Health in Tech Survey Dataset”, containing survey responses related to mental health.

Key Features Include:

- Age, Gender
- Country and workplace details
- Family history of mental illness
- Work interference due to stress
- Availability of mental health benefits
- Leave policies and support systems

4. Data Preprocessing

To ensure data quality and model efficiency, the following preprocessing steps were performed:

- Removal of irrelevant columns such as timestamp, comments, and state
- Handling missing values:
    - Categorical data — filled using mode
    - Numerical data — filled using median
- Encoding categorical variables using Label Encoding
- Feature scaling using StandardScaler
- Splitting dataset into:
    - 80% training data
    - 20% testing data

5. Methodology

This project uses supervised machine learning algorithms for classification.

Models Implemented:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest
- Gradient Boosting

Among these, Gradient Boosting and Random Forest are ensemble techniques that combine multiple learners to improve prediction performance.

6. Model Evaluation Metrics

The models were evaluated using:

- **Accuracy** — Overall correctness of predictions
- **Precision** — Correct positive predictions
- **F1 Score** —Balance between precision and recall
- **Confusion Matrix** —Visualization of classification results

7. Results and Findings

After training and evaluating all models, the following results were obtained:

Model Performance Summary:

Model	Accuracy (%)	F1 Score
Logistic Regression	71.82%	0.71
SVM	72.22%	0.72
Random Forest	75.00%	0.74
Gradient Boosting	75.39%	0.75

Key Observations:

- Gradient Boosting achieved the highest accuracy and F1 score
- Random Forest also performed well, closely matching Gradient Boosting
- Logistic Regression and SVM showed comparatively lower performance
- Ensemble models outperformed traditional models

8. Feature Importance Analysis

Feature importance was analyzed using the Random Forest model.

Top Influential Features:

- Work interference due to mental health
- Family history of mental illness
- Availability of workplace mental health care
- Gender
- Organization size (number of employees)
- Country
- Leave policies
- Employee benefits

9. Tools and Technologies

- **Programming Language:** Python
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Machine Learning Techniques:** Supervised Learning Algorithms
- **Development Environment:** Jupyter Notebook / Google Colab
- **Dataset Format:** CSV
- **Version Control (optional):** GitHub

10. Actual Code Snippet and Result

The following visual outputs were generated:

```python
def evaluate(y_test, y_pred, name):
    print("\nModel:", name)

    acc = accuracy_score(y_test, y_pred) * 100  # convert to %
    prec = precision_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    print(f"Accuracy: {acc:.2f}%")
    print(f"Precision: {prec:.2f}")
    print(f"F1 Score: {f1:.2f}")

evaluate(y_test, y_pred_lr, "Logistic Regression")
evaluate(y_test, y_pred_svm, "SVM")
evaluate(y_test, y_pred_rf, "Random Forest")
evaluate(y_test, y_pred_gb, "Gradient Boosting")
```

[13] Python

```
Model: Logistic Regression
Accuracy: 71.83%
Precision: 0.71
F1 Score: 0.71

Model: SVM
Accuracy: 72.22%
Precision: 0.70
F1 Score: 0.72

Model: Random Forest
Accuracy: 75.00%
Precision: 0.73
F1 Score: 0.75

Model: Gradient Boosting
Accuracy: 75.40%
Precision: 0.74
F1 Score: 0.75
```

This shows the model's accuracy , precision and f1 score

```python
importances = rf.feature_importances_
features = X.columns

importance_df = pd.DataFrame({
    "Feature": features,
    "Importance": importances
}).sort_values(by="Importance", ascending=False)

importance_df.head(10)
```

[14] Python

	Feature	Importance
5	work_interfere	0.125645
4	family_history	0.097724
10	care_options	0.071898
1	Gender	0.066090
6	no_employees	0.059019
2	Country	0.056823
14	leave	0.046587
9	benefits	0.042616
0	Age	0.042613
20	phys_health_interview	0.039429

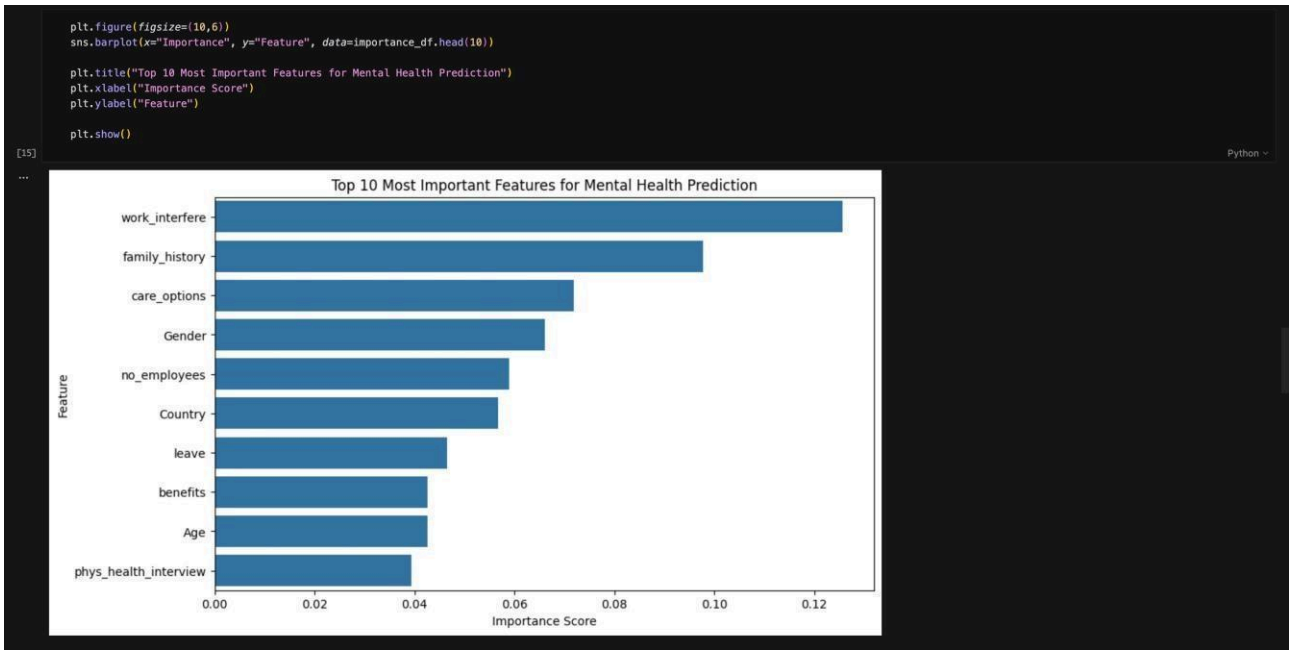

Figure 1: Feature Importance Bar Graph

Figure 2: Confusion Matrix for Gradient Boosting

```python
models = ["Logistic Regression", "SVM", "Random Forest", "Gradient Boosting"]

accuracy_scores = [
    accuracy_score(y_test, y_pred_lr),
    accuracy_score(y_test, y_pred_svm),
    accuracy_score(y_test, y_pred_rf),
    accuracy_score(y_test, y_pred_gb)
]

# Convert to percentage
accuracy_percent = [x * 100 for x in accuracy_scores]

plt.figure(figsize=(8,5))
sns.barplot(x=models, y=accuracy_percent)

plt.title("Model Accuracy Comparison")
plt.xlabel("Models")
plt.ylabel("Accuracy (%)")

plt.ylim(0,100)

plt.gca().yaxis.set_major_locator(mticker.MaxNLocator(integer=True))

plt.show()
```

[20] Python

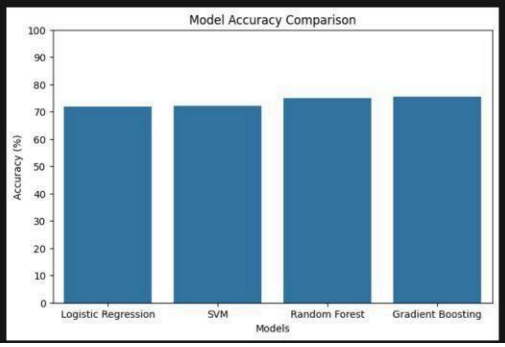

Figure 3: Model Accuracy Comparison Graph

```python
f1_scores = [
    f1_score(y_test, y_pred_lr),
    f1_score(y_test, y_pred_svm),
    f1_score(y_test, y_pred_rf),
    f1_score(y_test, y_pred_gb)
]

plt.figure(figsize=(8,5))
sns.barplot(x=models, y=f1_scores)

plt.title("Model F1 Score Comparison")
plt.xlabel("Machine Learning Models")
plt.ylabel("F1 Score")

plt.ylim(0,1)

plt.show()
```

[21] Python

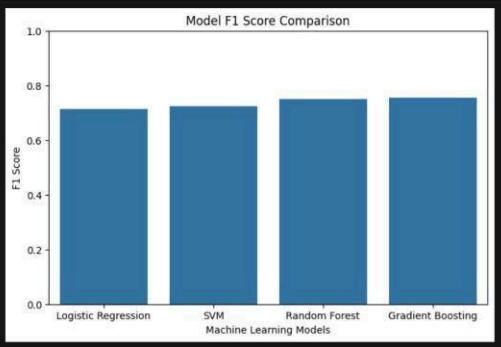

Figure 4: Model F1 Score Comparison Graph

11. Prediction Outcome

The final model predicts whether an individual is likely to need mental health treatment based on input features.

This prediction can be used as:

- An early warning system
- A decision-support tool
- A screening mechanism in workplaces or institutions

12. Conclusion

This project demonstrates that machine learning can effectively assist in the early prediction of mental health treatment needs. Among all models, Gradient Boosting performed the best, followed closely by Random Forest.

The findings highlight the importance of workplace environment and personal history in mental health outcomes. While the model cannot replace clinical diagnosis, it serves as a valuable tool for early detection and awareness.

13. Future Scope

- Use larger and more diverse datasets
- Apply deep learning techniques
- Integrate real-time behavioral or wearable data
- Improve model accuracy with hyperparameter tuning
- Ensure ethical considerations like data privacy and fairness

14. References

1. Dataset Source – Kaggle
2. Scikit-learn Documentation
3. Research articles on machine learning in mental health
4. World Health Organization (WHO) reports on mental health

15. Appendices

Appendix A: Dataset Description
The dataset used in this study consists of structured data in CSV format, containing relevant features required for prediction and analysis. It includes both input variables and the target variable used for model training and evaluation.

Appendix B: Model Implementation
The model was implemented using Python and standard machine learning libraries. Data preprocessing steps such as handling missing values, normalization, and feature selection were performed before training.

Appendix C: Evaluation Metrics
Performance of the model was evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliable results.